Lecture 5
Distributions, Normal Model

Part 2.

THE NORMAL MODEL

UNLEASH
TOMORROW

The Normal Model

The normal distribution is one of the most widely used continuous probability distributions.

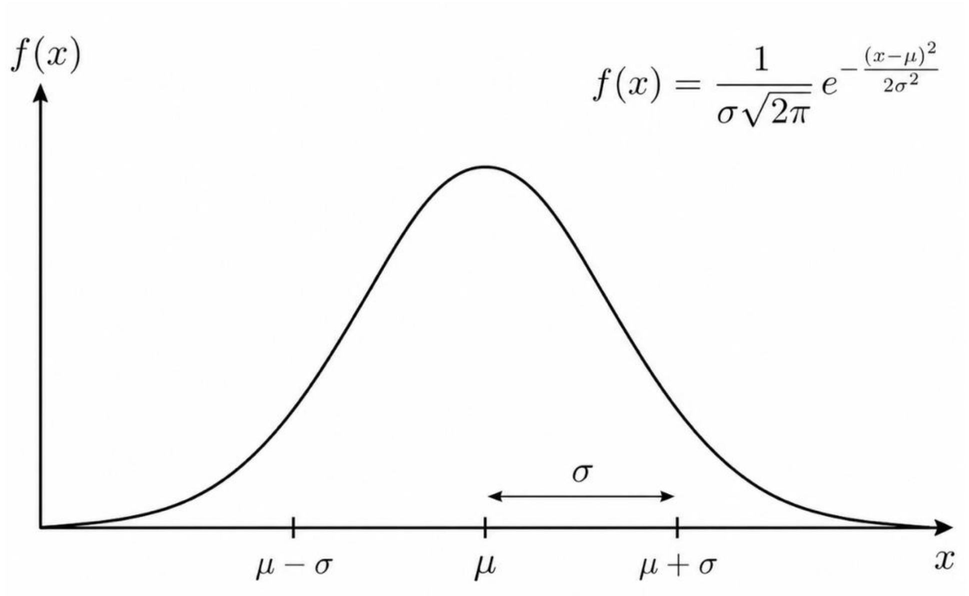
It models variables that tend to be concentrated around a central value, with values further from the center becoming less likely.

The curve is:

- symmetric,
- centered at μ,
- controlled in spread by σ,
- with total area equal to 1.

We typically write : $X \sim \mathcal{N}(\mu, \sigma^2)$

This should be read as:
The continuous random variable X follows a normal distribution with mean μ and standard deviation σ^2.

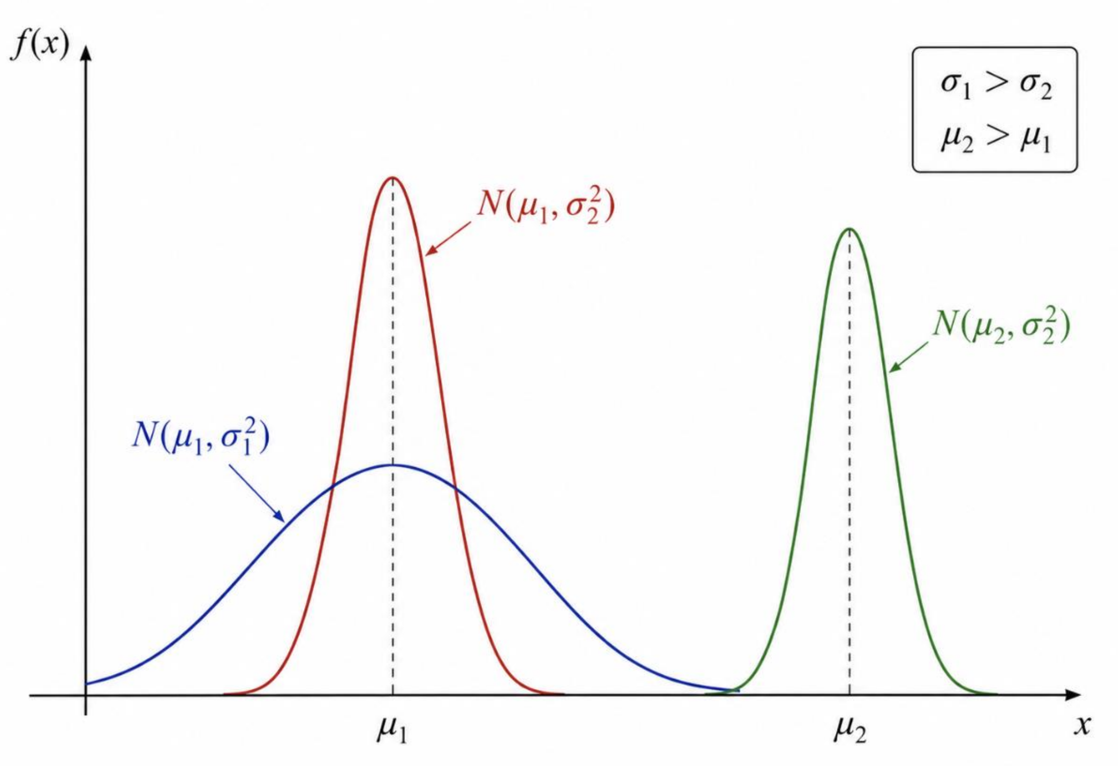


Mean and Standard Deviation

A normal distribution is determined by two parameters.

$$\mu = mean$$
$$\sigma = standard\ deviation$$

- The mean μ determines the center of the distribution.
- The standard deviation σ determines how dispersed the values are.
- A small σ gives a narrow curve.
- A large σ gives a wider curve.



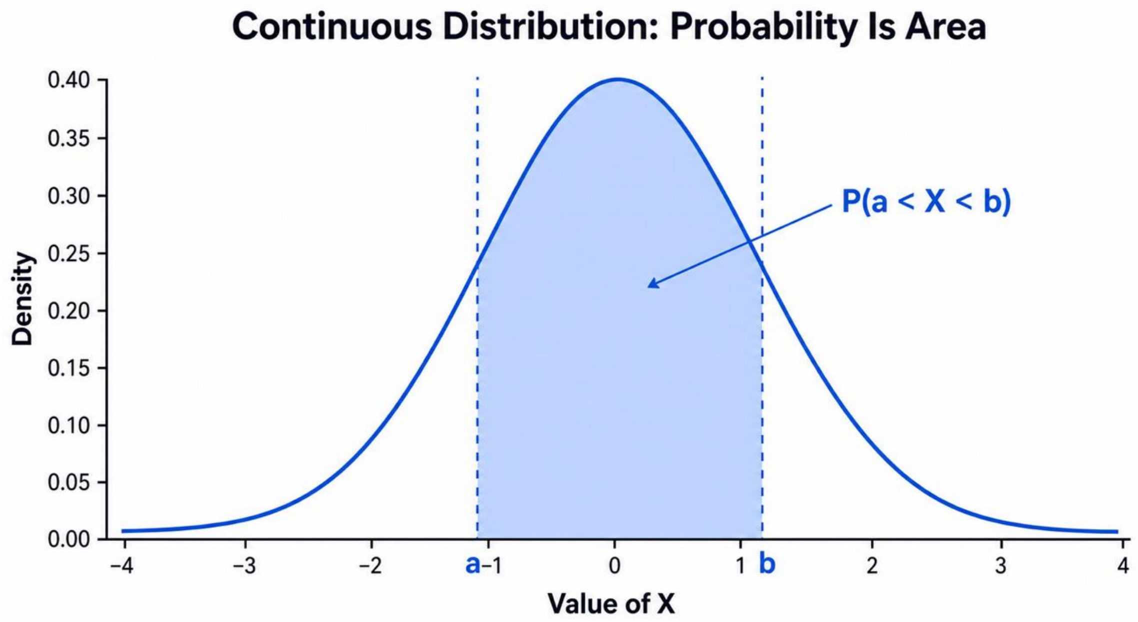
Probabilities Under the Normal Curve

For a normal random variable, probabilities are areas under the density curve.

$$P(a < X < b)$$

means the area under the curve between a and b.

Unlike the uniform distribution, this area cannot be computed with a simple rectangle formula.



Cumulative Distribution Function

Let $F(x)$ denote the cumulative distribution function (CDF) of the normal distribution:

$$F(x) = P(X \leq x)$$

That is, $F(x)$ gives the probability that the random variable X takes a value less than or equal to x.

Therefore,

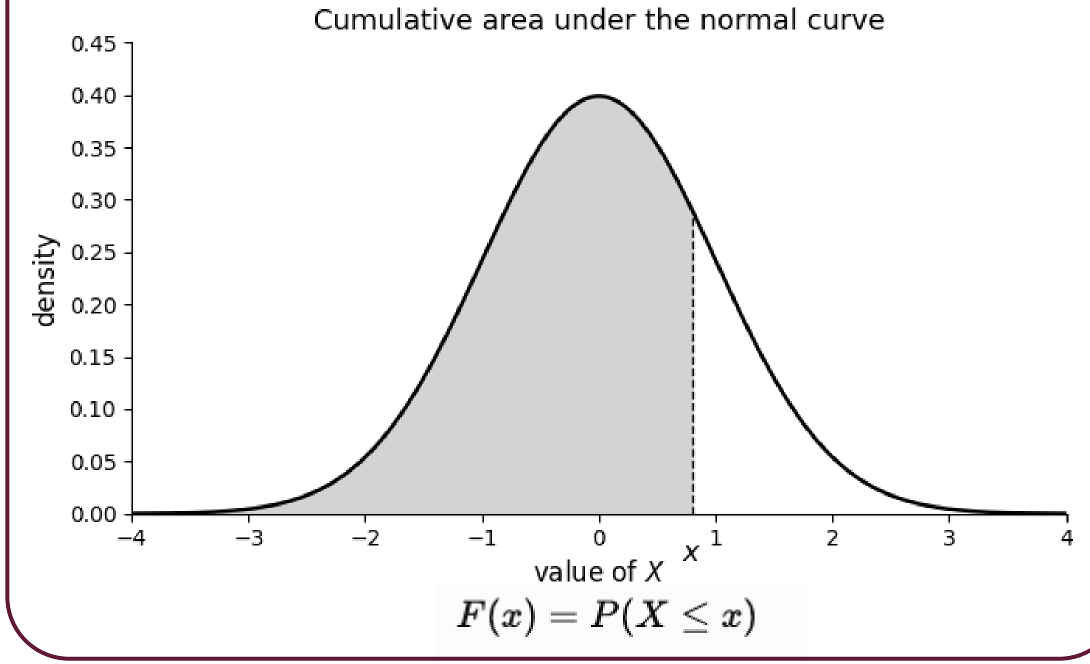
$$F(a) = P(X \leq a)$$

and

$$F(b) = P(X \leq b).$$

Since the probability up to b includes the probability up to a, the probability that X lies between a and b is simply the difference:

$$\boxed{P(a \leq X \leq b) = F(b) - F(a).}$$



Applied example

Suppose X represents the daily demand for a product.
Assume demand is normally distributed with:

$$\mu = 100$$
$$\sigma = 15$$

We write:

$$X \sim \mathcal{N}(100, 15^2)$$

Question:

$$P(90 < X < 120)$$

What is the probability that daily demand is between 90 and 120 units?

Computing probabilities in practice

The function `norm.cdf(x, loc=mu, scale=sigma)` computes:

$$P(X \leq x)$$

So:

$$P(90 < X < 120) = F(120) - F(90)$$

```
from scipy.stats import norm

mu = 100
sigma = 15

prob = norm.cdf(120, loc=mu, scale=sigma) - norm.cdf(90, loc=mu, scale=sigma)

print(prob)
```

```
>> 0.6562962427272092
```

Standardization

Today, statistical software can compute probabilities directly for any normal distribution. Historically, however, this was not possible, so every normal variable was first transformed into a standard normal variable.

$$Z = \frac{X - \mu}{\sigma}$$

The variable Z follows the standard normal distribution:

$$Z \sim \mathcal{N}(0, 1)$$

This transformation is called standardization.

Although Python can compute probabilities directly, standardization remains important because it provides a universal way of comparing values and was historically the method used with printed normal probability tables.

z-score

A z-score tells us how many standard deviations a value is above or below the mean.

$$z = \frac{x - \mu}{\sigma}$$

Example:

$$\mu = 100,$$
$$\sigma = 15,$$
$$x = 120$$

$$z = \frac{120 - 100}{15} = 1.33$$

So the value x = 120 is 1.33 standard deviations above the mean.

Direct Computation vs Standardization

We can compute normal probabilities directly:

$$P(X < 120)$$

or by standardizing:

$$P(X < 120) = P\left(Z < \frac{120 - \mu}{\sigma}\right)$$

For $X \sim \mathcal{N}(100, 15^2)$:

$$P(X < 120) = P\left(Z < \frac{120 - 100}{15}\right)$$

$$P(X < 120) = P(Z < 1.33)$$

Both approaches give the same probability.

From Data to a Distribution

The process is:

data → *histogram* → *fitted distribution*

A fitted distribution is a theoretical model chosen to approximate the empirical data.
For a normal model, we estimate:

$$\mu \approx \bar{x}$$
$$\sigma \approx s$$

where $\bar{x}$ is the sample mean and $\boldsymbol{s}$ is the sample standard deviation.

Practice: Fitting real data to normal density function

We use the file: **daily log returns 2015 2026.csv**

The variable of interest is:

$$X = \textit{daily log return}$$

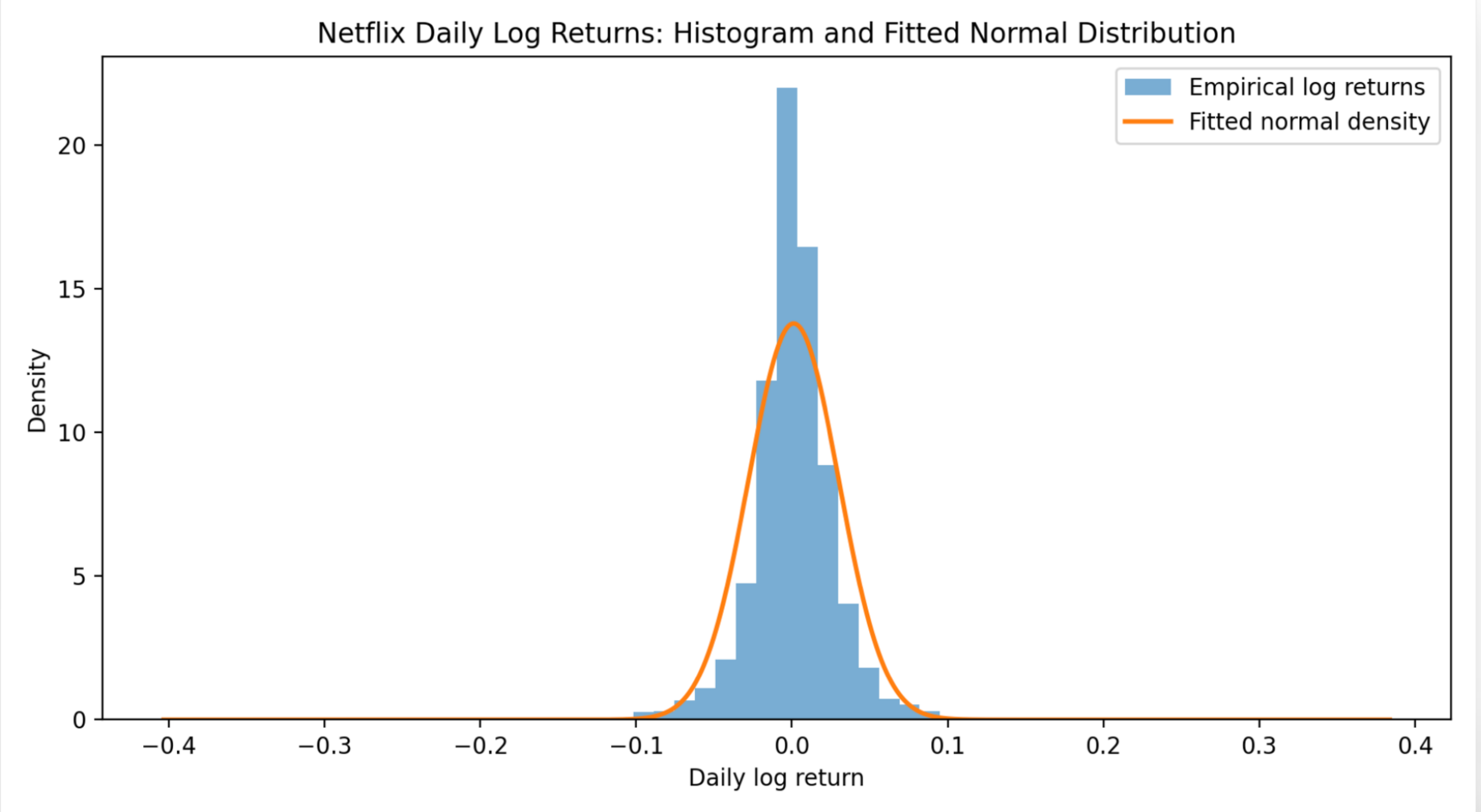
Goal: fit the data and test visually if the normal distribution is a good model

$$X \sim \mathcal{N}(\mu, \sigma^2)$$

We will use python to

1. read the data;
2. compute μ, σ^2 ,and σ;
3. compare the histogram with the fitted normal density;
4. compare theoretical and empirical tail probabilities.

Parameter estimation



Estimated parameters:

$$\mu \approx 0.00109$$
$$\sigma^2 \approx 0.000837$$
$$\sigma \approx 0.02894$$

Therefore:

$$X \sim \mathcal{N}(0.00109, 0.000837)$$

See video also in this lecture about fitting the normal distribution with python.

Does the Normal Model Match Reality?

The normal model gives us theoretical probabilities. The data give us empirical probabilities. Comparing the two tells us where the model is a good approximation and where it may be misleading.

Business question	Probability question	Empirical data	Normal model
Daily log return between -2% and $+2\%$	$P(-0.02 < X < 0.02)$	68.0%	51.0%
Daily log return between $-5\%$and -2%	$P(-0.05 < X < -0.02)$	11.5%	19.4%
Daily log return lower than -3%	$P(X < -0.03)$	8.5%	14.1%
Daily log return higher than $+4\%$	$P(X > 0.04)$	5.4%	8.9%
Daily log return more than 3σ below the mean	$P(X < \mu - 3\sigma)$	0.70%	0.135%

Takeaway:
The normal model is useful as an approximation, but it underestimates very extreme negative returns in this dataset.

more
less

Limits of the Normal Model

The normal model is powerful, but it is not always appropriate.

It may be problematic when the data are:

- ❑ strongly skewed,
- ❑ bounded below by zero,
- ❑ dominated by extreme values,
- ❑ clearly multi-modal,
- ❑ affected by rare but large shocks.

Example: financial returns may look approximately normal in the center, but often have **heavier tails** than the normal distribution,